

PerfSight Lite Documentation

Version: 1.0.0

Publisher: JGT Tools

Introduction

Thank you for purchasing/downloading PerfSight Lite.

PerfSight Lite is a lightweight runtime performance overlay for Unity that provides developers with real-time visibility into key performance metrics and system information during development and testing.

The tool is designed to be quick to integrate, easy to use, and suitable for both Editor and standalone build environments.

Features

Performance Metrics

- Frames Per Second (FPS)
- Frame Time (ms)
- Memory Usage
- Garbage Collection Allocations
- Draw Calls
- Triangle Count
- Vertex Count

System Information

- CPU
- GPU
- RAM
- Graphics API
- Driver Information
- Operating System
- Screen Resolution

Runtime Support

- Unity Editor
 - Play Mode
 - Standalone Builds
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Requirements

Supported Unity Versions

- Unity 2021.3.45f2 LTS or newer
- Unity 6.x

Supported Platforms

- Windows
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Installation

Importing the Package

1. Import the PerfSight Lite package into your project.
2. Wait for Unity to complete asset import.
3. Locate the PerfSight folder within your Project window.

Example: *Assets/JGT Tools/PerfSight Lite/*

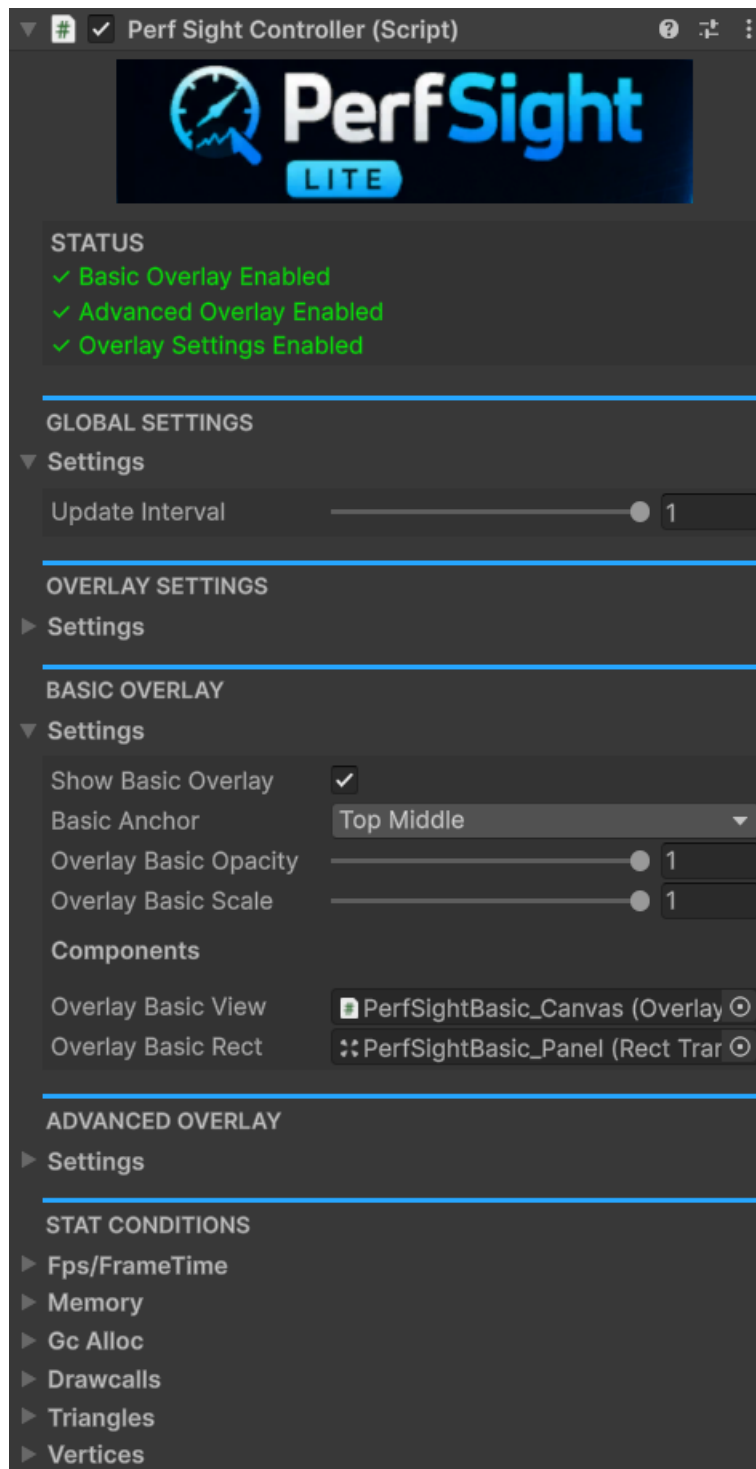
Quick Start

Adding PerfSight To Your Scene

1. Open the desired scene.
2. Drag the PerfSight-Lite prefab into the scene.
3. Enter Play Mode.

The overlay should now appear automatically.

Further configuration can be achieved by modifying the values exposed inside the Perf Sight Controller component.



Understanding The Metrics

FPS

Frames rendered per second.

Higher values generally indicate smoother performance.

Frame Time

Time required to render a frame.

Measured in milliseconds.

Lower values indicate better performance.

Memory

Current memory usage reported by Unity.

GC Alloc

Garbage collection allocations generated during runtime.

Frequent allocations may lead to performance spikes.

Note: There is significant GC Alloc overhead inside the editor, a buffer value is used to account for this as per my testing. This buffer value is accessible inside the PerfSightController component if you wish to modify it.

Draw Calls

Number of rendering submissions issued to the graphics API.

Reducing draw calls can improve rendering performance.

Triangles

Total triangles rendered during the frame.

Useful when diagnosing rendering complexity.

Vertices

Total vertices processed during rendering.

Useful for identifying expensive geometry.

Performance Considerations

PerfSight Lite is designed to remain lightweight during operation.

Recommended practices:

- Use build testing for final performance validation.
 - Avoid relying solely on editor performance measurements.
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Troubleshooting

Overlay Not Visible

Check:

- The PerfSight prefab exists in the scene and is active.
 - The Canvases are active.
 - Check canvas sort ordering values based on your own setup
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Incorrect Values

Some metrics may vary depending on:

- Unity version
- Render Pipeline
- Platform
- Build configuration

Always validate performance using a development build when possible.

Build Issues

If PerfSight does not appear in a build:

1. Verify the prefab/component exists in the startup scene.
 2. Confirm no compilation errors are present.
 3. Ensure stripping settings are not removing required assemblies.
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Frequently Asked Questions

Does PerfSight replace the Unity Profiler?

No.

PerfSight is intended as a lightweight runtime monitoring tool that complements the Unity Profiler.

Can I use PerfSight in shipped builds?

Yes.

However, it is primarily intended for development and testing workflows.

Can I reposition PerfSight overlays in builds?

Yes.

Make sure the Overlay Settings is enabled in the PerfSightController component before compiling your build. Then you can use this to reposition, enable/disable and change opacity and scale.

Changelog

Version 1.0

Initial release.

Features:

- FPS
 - Frame Time
 - Memory Usage
 - GC Allocations
 - Draw Calls
 - Triangle Count
 - Vertex Count
 - System Information
 - Multiple configurable overlays
 - Can be used in Editor and Standalone builds
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Support

Publisher: JGT Tools

For bug reports, feature requests, and support inquiries:

Email: jgt-tools@outlook.com

Website: <https://jgt-tools.loveable.app>

Thank you for using PerfSight Lite.